



BAULKHAM HILLS HIGH SCHOOL

2016

YEAR 11 December - Task 1

Mathematics Extension 2

General Instructions

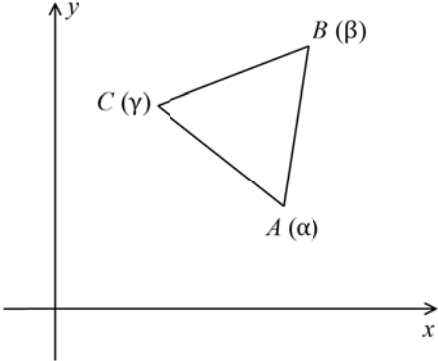
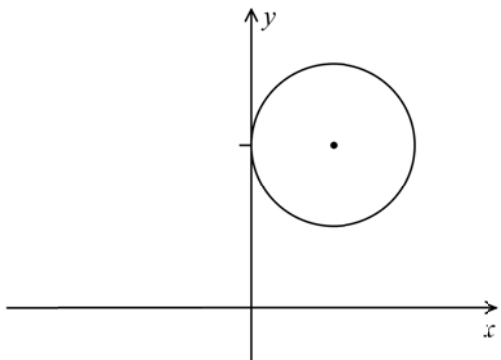
- Reading time – 5 minutes
- Working time – 50 minutes
- Write using non erasable black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

Total marks – 35

Exam consists of 3 pages.

Question 1 (11 marks) Start on the appropriate page of your answer booklet.		Marks
a)	Find :	
	(i) $ 2 + 3i $	1
	(ii) $\arg(-1 - i)$	1
	(iii) $\frac{4+3i}{2-i}$	2
b)	(i) Find $\sqrt{15 - 8i}$	2
	(ii) Hence solve $z^2 - (2 + 3i)z - 5 + 5i = 0$	3
c)	Draw a neat sketch of the following locus: $z\bar{z} - 3(z + \bar{z}) \leq 0$	2

Question 2 (12 marks) Start on the appropriate page of your answer booklet.		Marks
a)	(i) If $w = \frac{1-i\sqrt{3}}{2}$, show $w^3 = -1$.	2
	(ii) Hence calculate w^{11} .	2
b)	If $z = \cos \theta + i \sin \theta$	
	i) Given that $(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$ show that $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$.	3
	ii) Hence show that polynomial equation $16x^5 - 20x^3 + 5x - 1 = 0$ has roots $\cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}$ and 1.	2
	iii) Hence deduce that $\cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} = -\frac{1}{4}$.	3

Question 3 (12 marks) Start on the appropriate page of your answer booklet.	Marks
a) Find the equation of the locus of z, stating any restrictions, such that: $\arg \left[\frac{z - 1 + 2i}{z - 3 + 2i} \right] = \frac{\pi}{4}$	3
b) Let $w = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ (i) Show that $w^3 = 1$. (ii) Show that $1 + w + w^2 = 0$. The points A, B and C form an equilateral triangle in the Argand diagram, and represent the complex numbers α, β, γ respectively.  (iii) Show that $\alpha - \gamma = w(\gamma - \beta)$. (iv) Deduce that $\alpha + w\beta + w^2\gamma = 0$.	1 1 2 1
c) The locus of z, such that $z - 1 - 2i = 1$ is sketched below. 	
(i) Find the maximum value of $z + 1$.	2
(ii) Find the maximum value of $\arg(z + 1)$ correct to the nearest degree.	2

End of Exam

X2 - solutions - Dec. 2016

Question 1 (11 marks)

a) Find

(i) $|2+3i| = \sqrt{2^2+3^2} = \sqrt{13}$

(ii) $\text{Arg}(-1-i) = -\frac{3\pi}{4}$

iii) $\frac{4+3i}{2-i} \times \frac{2+i}{2+i} = \frac{8-3+ i(6+4)}{4+1}$

$= \frac{5+10i}{5} = 1+2i$

Marks

1 - correct solns

1 - correct solns

2 - correct solns

1 - multiplies by conjugate

b) i) Let $\sqrt{15-8i} = z$ where $z = x+iy$

$\therefore 15-8i = z^2 = x^2-y^2+2xyi$

Compare real & imaginary parts

$\therefore (1) \quad 15 = x^2-y^2$

$(2) \quad -8 = 2xy \therefore x = -\frac{4}{y}$

$\therefore 15 = \left(-\frac{4}{y}\right)^2 - y^2$

$\therefore y^4 + 15y^2 - 16 = 0$

$(y^2+16)(y^2-1) = 0$

$y^2+16 = 0 \quad y = \pm 1$
no soln.

for y -real

$\therefore y=1 \therefore x=-4$
 $y=-1 \therefore x=4$ } $\therefore \sqrt{15-8i} = \begin{matrix} -4+i \\ 4-i \end{matrix}$

or $\sqrt{15-8i} = \pm (4-i)$

2 - correct solns

1 - attempts to solve simult. equations by equating real & imaginary parts

Question 1 b) cont. (3 marks)

Marks

ii) $z^2 - (2+3i)z - 5+5i = 0$

$$z = \frac{2+3i \pm \sqrt{[-(2+3i)]^2 - 4 \times (-5+5i)}}{2}$$

3 - correct solns.

$$z = \frac{2+3i \pm \sqrt{4+12i-9+20-20i}}{2}$$

2 - manipulates quad. formula

$$z = \frac{2+3i \pm \sqrt{15-8i}}{2}$$

to achieve $\Delta = 15-8i$

from part i) $z = \frac{2+3i \pm (\pm(4-i))}{2}$

1 - substitutes correctly into quadratic formula

$$\therefore z = \frac{2+3i \pm (4-i)}{2}$$

$$z = \frac{6+2i}{2} = 3+i$$

$$z = \frac{-2+4i}{2} = -1+2i$$

Question 1c) (2 marks)

$$z\bar{z} - 3(z + \bar{z}) \leq 0$$

Marks

$$\begin{aligned} \text{let } z &= x + iy & z\bar{z} &= x^2 + y^2 \\ z + \bar{z} &= 2\operatorname{Re}(z) = 2x \end{aligned}$$

2 - correct
sketch

$$\therefore x^2 + y^2 - 3(2x) \leq 0$$

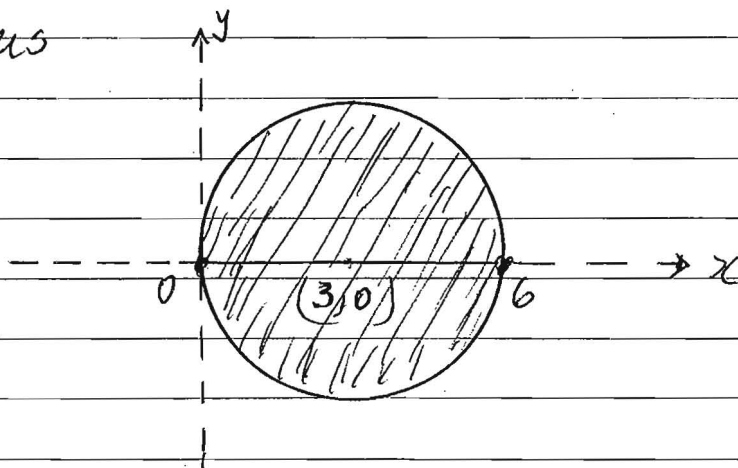
$$\therefore x^2 - 6x + 9 + y^2 \leq 0 + 9$$

$$\therefore (x - 3)^2 + y^2 \leq 9$$

1 - finding equation
of the circle

$\therefore$ boundary is a circle centre (3, 0)
radius 3

locus



Question 2 (12 marks)

a) i) if $w = \frac{1-i\sqrt{3}}{2} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$

$$w = |w| \operatorname{cis} \theta$$

$$|w| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2} = 1$$

$$\theta = \tan^{-1} \frac{\sqrt{3}/2}{1/2} = -\frac{\pi}{3}$$

$$\therefore w = 1 \times \operatorname{cis}\left(-\frac{\pi}{3}\right)$$

$$\therefore w^3 = 1^3 \operatorname{cis}\left[3 \times -\frac{\pi}{3}\right]$$

$$= \operatorname{cis}(-\pi) = -1 \text{ as shown.}$$

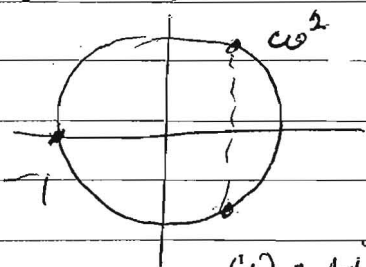
ii) $w'' = (w^3)^3 \times w^2$

from (i) $w'' = (-1)^3 \times w^2 = -w^2 = -\operatorname{cis}\left(-\frac{2\pi}{3}\right)$

$$w'' = +\operatorname{cis} \frac{\pi}{3} \text{ or}$$

$$w'' = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(OR) since w is one of the complex roots of $w^3 = -1$



$$w = \operatorname{cis}\left(-\frac{\pi}{3}\right)$$

$\therefore$ the other roots are $w = -1$

$$\text{and } w^2 = \bar{w} = \operatorname{cis} \frac{\pi}{3}$$

$$\therefore w'' = w^2 = \operatorname{cis} \frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

Marks

2 - correct working

1 - expressing w in mod/arg. form

2 - correct solns.

1 - manipulates

$w'' = (w^3)^3 \times w^2$ correctly

Question 2 - cont.

26) i) If $z = \cos \theta + i \sin \theta$

$$\text{show } \cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$

since $z = \cos \theta + i \sin \theta$

$$\therefore z^5 = (\cos \theta + i \sin \theta)^5 \neq z^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{Now: } (\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5\cos^4 \theta i \sin \theta + 10\cos^3 \theta i^2 \sin^2 \theta \\ + 10\cos^2 \theta i^3 \sin^3 \theta + 5\cos \theta i^4 \sin^4 \theta + i^5 \sin^5 \theta$$

and since $z^5 = \cos 5\theta + i \sin 5\theta$ (De Moivre's Theorem)

Now equate real parts of both expansions

$$\therefore \cos 5\theta = \cos^5 \theta + 10\cos^3 \theta i^2 \sin^2 \theta + 5\cos \theta i^4 \sin^4 \theta$$

$$\therefore \cos 5\theta = \cos^5 \theta - 10\cos^3 \theta \sin^2 \theta + 5\cos \theta \sin^4 \theta$$

$$= \cos^5 \theta - 10\cos^3 \theta (1 - \cos^2 \theta) + 5\cos \theta (1 - \cos^2 \theta)^2$$

$$= \cos^5 \theta - 10\cos^3 \theta + 10\cos^5 \theta + 5\cos \theta - 10\cos^3 \theta + 5\cos^5 \theta$$

$$\therefore \cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta \therefore \text{shown}$$

Marks

3 - correct working

2 - correctly uses De Moivre's Theorem and equates real parts

1 - expands binomial exp. and simplifies correctly

Question 2 b) cont.

Marks

(ii) let $x = \cos \theta$

since $16x^5 - 20x^3 + 5x - 1 = 0$

$$\therefore 16x^5 - 20x^3 + 5x = 1$$

which is $16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = 1$

2 - correct explanation

1 - solves $\cos 5\theta = 1$ for θ .

from i) $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$

$\therefore$ equation to solve $\cos 5\theta = 1$

so the roots of $16x^5 - 20x^3 + 5x - 1 = 0$

are the same as roots of $\cos 5\theta = 1$

which are $\therefore 5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$

$\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$

but $x = \cos \theta \therefore x = 1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}$

and $\cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}$

$\therefore$ shown

(iii) Deduce $\cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} = -\frac{1}{4}$

3 - correct explanation

solns: From ii) Roots of $16x^5 - 20x^3 + 5x - 1 = 0$

are $\alpha = 1, \beta = \cos \frac{2\pi}{5}, \gamma = \cos \frac{4\pi}{5}, \delta = \cos \frac{6\pi}{5}, \epsilon = \cos \frac{8\pi}{5}$

2 - expresses the product of roots in terms of $\cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5}$

Now product of roots

$$\alpha \beta \gamma \delta \epsilon = -\frac{c}{a} = -\frac{-1}{16}$$

with correct justification

$$\therefore 1 \times \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} \times \cos \frac{6\pi}{5} \times \cos \frac{8\pi}{5} = \frac{1}{16}$$

1 - calculates the product of roots of eqn. from ii)

but $\cos \frac{2\pi}{5} = \cos \left(2\pi - \frac{2\pi}{5} \right) = \cos \frac{8\pi}{5}$

and $\cos \frac{4\pi}{5} = \cos \left(2\pi - \frac{4\pi}{5} \right) = \cos \frac{6\pi}{5}$

$\therefore 1 \times \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} \times \cos \frac{4\pi}{5} \times \cos \frac{2\pi}{5} = \frac{1}{16}$

PTD

Q.2 b ii) cont.

$$\therefore \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} = \frac{1}{16}$$

$$\therefore \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} = \pm \sqrt{\frac{1}{16}} = \pm \frac{1}{4}$$

but since $\cos \frac{2\pi}{5} > 0$ and $\cos \frac{4\pi}{5} < 0$

$$\therefore \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} < 0$$

$$\therefore \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} = -\frac{1}{4}$$

Question 3 (12 marks)

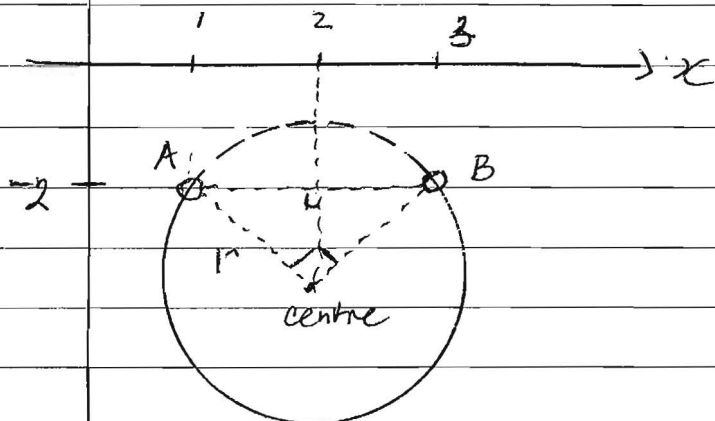
Marks

$$a) \arg \left[\frac{z-1+2i}{z-3+2i} \right] = \arg \left[\frac{z-(1-2i)}{z-(3-2i)} \right] = \frac{\pi}{4}$$

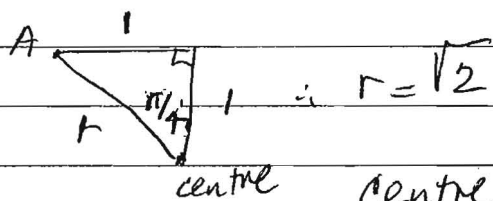
3-correct solns.

let $A(1, -2)$ and $B(3, -2)$

2-finds the correct equation of circle



1-correct sketch with open circles at $(1, -2)$ & $(3, -2)$



centre $(2, -2-1) = (2, -3)$

equation of the circle (A+B)

$$(x-2)^2 + (y+3)^2 = 2$$

$\therefore$ Locus of z is the major arc of this circle excluding $A(1, -2)$ & $B(3, -2)$

Now with restrictions

it's a part of circle $(x-2)^2 + (y+3)^2 = 2$
where $y < -2$

(or) circle $(x-2)^2 + (y+3)^2 = 2$
where $-3-\sqrt{2} \leq y < -2$

Question 3 b)

Let $w = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

i) $w^3 = \cos\left(3 \times \frac{2\pi}{3}\right) + i \sin\left(3 \times \frac{2\pi}{3}\right)$

$= \cos 2\pi + i \sin 2\pi = +1$

Marks

1 - correct explanation

ii) Now since $w^3 = 1$

$\therefore w^3 - 1 = 0$

$\therefore (w-1)(w^2 + w + 1) = 0$

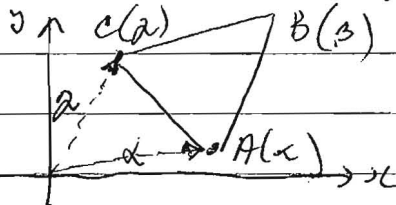
but $w \neq 1$
since it's a complex root $\therefore w^2 + w + 1 = 0$

1 - correct solns with reasoning

(OR) by sum of roots of $w^3 - 1 = 0$

(OR) manipulates $1 + w + w^2 = 1 + \text{cis } \frac{2\pi}{3} + \text{cis } \frac{4\pi}{3} = \dots$

(iii) Show that $\alpha - \beta = w(\beta - \gamma)$.



since ΔABC is equilateral $\therefore |\vec{AB}| = |\vec{BC}| = |\vec{AC}|$

Now $\alpha - \beta = \vec{CA}$ and $\beta - \gamma = \vec{BC}$

Since $w = \text{cis } \frac{2\pi}{3}$ and $|w| = 1$

$\therefore w(\beta - \gamma) = w \times \vec{BC} = \vec{CA}$

which is rotating $\vec{BC}$ by $\frac{2\pi}{3}$ anticlockwise

2 - correct working

1 - progress towards solns. by rotating vectors

- manipulates

$\alpha - \beta = \vec{CA}$
or $\beta - \gamma = \vec{BC}$
by multiplying by w

(OR) $\alpha - \beta = \vec{CA} = \vec{CB} \times \text{cis}\left(-\frac{\pi}{3}\right)$

$= \vec{BC} \times \text{cis}\pi \times \text{cis}\left(-\frac{\pi}{3}\right)$

$= \vec{BC} \times \text{cis}\left(\frac{2\pi}{3}\right) = \vec{BC} \times w = (\beta - \gamma) \times w \therefore \text{proven}$

3b) iv)

Deduce $\alpha + \omega\beta + \omega^2\gamma = 0$

Marks

Solns: From iii) $\alpha - \gamma = \omega(\gamma - \beta)$

1 - correct solution

$\therefore \alpha - \gamma = \omega\gamma - \omega\beta$

$\therefore \alpha - \gamma - \omega\gamma + \omega\beta = 0$

$\therefore \alpha + \omega\beta - \gamma(1 + \omega) = 0$

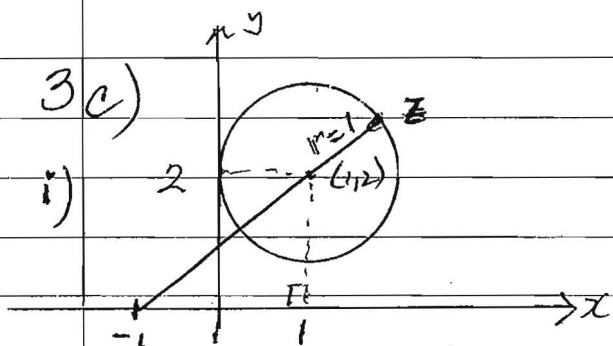
but from (ii) $1 + \omega + \omega^2 = 0 \therefore 1 + \omega = -\omega^2$

$\therefore \alpha + \omega\beta - \gamma(-\omega^2) = 0$

$\therefore \alpha + \omega\beta + \omega^2\gamma = 0 \therefore$ shown

3c)

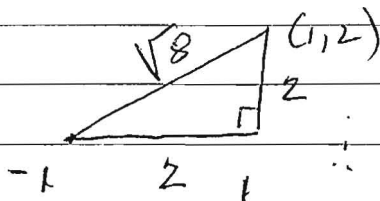
i)



2 - correct solution

The locus of $|z - (1 + 2i)| = 1$

is a circle centre $(1, 2)$, radius = 1



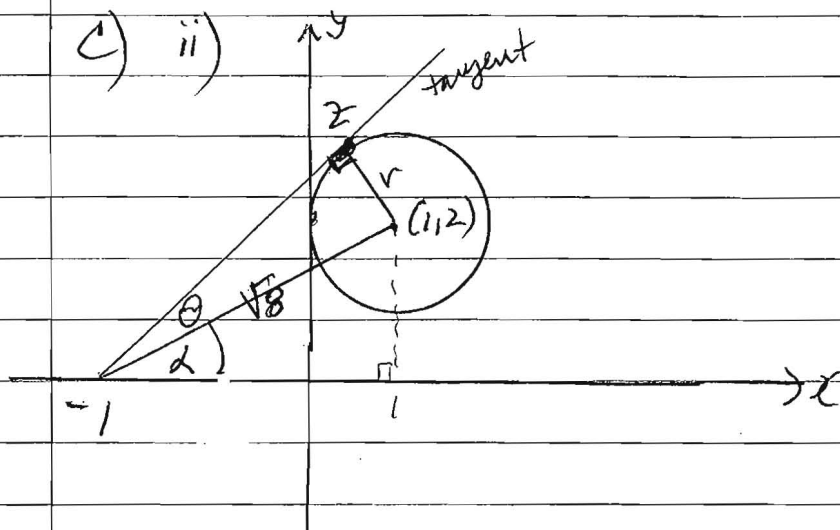
$\therefore$ Max. of $|z + 1| = \sqrt{8} + 1$

1 - finds distance from $(-1, 0)$ to centre of circle $(1, 2)$

- recognises z with Max. value $|z + 1|$ to be intersection of line joining $(-1, 0) \times (1, 2) \times$ circle

Question 3

c) ii)



Marks

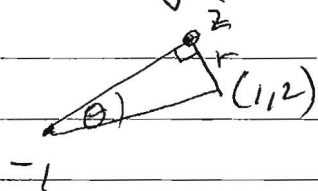
2 - correct solution

1 - finds arg. of vector of $(1, 2)$ from $(-1, 0)$

$$\alpha = \arg(1+2i - (-1))$$

1 - finds angle θ subtended by tangent from $(-1, 0)$ to circle and line joining the centre $(1, 2)$.

$$\alpha = \arg(1+2i - (-1)) = \tan^{-1} \frac{2}{2} = \frac{\pi}{4} = 45^\circ$$



$$\sin \theta = \frac{r}{\sqrt{8}} = \frac{1}{\sqrt{8}}$$

$$\therefore \theta = \sin^{-1} \frac{1}{\sqrt{8}} = 20.7^\circ$$

$$\therefore \text{max. value of } \arg(z+1) = 45^\circ + 20.7^\circ \\ \doteq 66^\circ \text{ (nearest degree)}$$